On the Quantization of the Dissipative Systems*

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March 10, 1948

Since our main interests consisted in the elementary processes, the quantization of the dissipative systems has escaped from our attack up to the present. But we find an example of such dissipative systems in the classical electron equations proposed by Dirac † which contain the so-called $\dddot{x}$-forces. So we shall here try to quantize such a system and investigate the results obtained.

First, we take up the ordinary dissipative system which does not involve the $\dddot{x}$-forces. If we confine our attention to the holonomic system, its Lagrangian equations of motion become as follows:

$$\frac{d\Pi_i}{dt}-\frac{\partial T}{\partial q_i}=-\frac{\partial V}{\partial q_i}+Q_i\,,$$

$$\Pi_i=\frac{\partial T}{\partial \dot{q}_i}\equiv\partial T/\partial\left(\frac{dq_i}{dt}\right)\,,$$

$$(i=1,\ 2,\ \cdots\cdots,\ n),$$

where q_i's are the generalized coordinates, T the kinetic energy and V the potential energy of this system. And $Q_i=Q_i\ (q,\ \dot{q},\ t)\ (i=1,2,\cdots\cdots,\ n)$ are the dissipative forces characterizing the present system.

In order to determine the necessary commutation relation:

$$[q_i,\ q_j]_-,\ [\Pi_i,\ \Pi_j]_-,\ [\Pi_i,\ q_j]_-$$
$$(i,\ j=1,\ 2,\cdots\cdots n),$$

we shall rely upon the following principles:

(A) Compatibility with the equations of motion,

(B) In the limit when all Q_i's tend to zero,

$$[q_i,q_j]_-=[\Pi_i,\Pi_j]_-=0\,,$$
$$[\Pi_i,\ q_j]_-=\frac{h}{i}\delta_{ij}\,.$$

As an example, we consider the simple harmonic oscillator subjected to the $\dot{x}$-resistance only in the one-dimensional motion. Then we have

$$T=\frac{1}{2}m\dot{x}^2,\ V=\frac{1}{2}m\omega^2x^2\,,$$
$$Q=-m\alpha\dot{x},\ (\alpha>0)$$

and the equations of motion are as follows:

$$\dot{x}=\frac{1}{m}\Pi,\ \dot{\Pi}=-m\omega^2x-\alpha\Pi\,.$$

Using the two principles (A) and (B), we can derive the next results with case:

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$$[p,\ q]_- = \frac{\hbar}{i}, \qquad p = e^{\alpha t}\Pi$$

and

$$i\hbar\dot{x} = [x,\ H]_-,\ i\hbar\dot{p} = [p,\ H]$$

where H may be interpreted as the Hamiltonian of this system:

$$H = \frac{1}{2m}e^{-\alpha t}p^2 + \frac{1}{2}m\omega^2 e^{\alpha t}x^2.$$

Since H depends on the time explicitly, we cannot have the stationary states at all. However, if we adopt the Schrödinger's description instead of the Heisenberg's one, we can show that there exist the pseudo-stationary states ψ_n $(n=0, 1, 2, \cdots)$ which turn into the true stationary states in the limit when $\alpha \to 0$ and are characterized by the quantum symbol E_n respectively,

$$E_n = \hbar\omega\sqrt{1-(\alpha/2\omega)^2}\ (n+1/2), \qquad (\alpha < 2\omega).$$

ψ_n gives the constant expectation value to H, but not to the classical energy. In fact

$$\langle 1/2m \cdot x^2 + 1/2m \cdot \omega^2 \dot{x}^2 : \ \exp \sim e^{\ \alpha t}.$$

Similarly, the amplitude of this oscillator decreases gradually as is expected by the classical mechanics. Generally we refind the well-known theorem that the expectation value of any dynamical variable behaves as though the corresponding classical quantity.

Next, we shall consider the case of the $\dddot{x}$-reaction force. If we confine us to the one dimensional motion subjected to the elastic binding, we can reduce this problem to the one above considered. Namely, the commutation relations $[x,\ \dot{x}]_-$, $[\dot{x},\ \ddot{x}]_-$ $[\dddot{x},\ x]$ can be determined by means of (A) and (B), and after some calculations we find the canonical equations:

$$i\hbar\dot{x} = [x,\ H]_-,\ i\hbar\dot{p} = [p,\ H]_-, \qquad p \equiv me^{\nu_0\gamma t}\dot{x},$$

where the Hamiltonian is as follows:

$$H = \frac{1}{2m}e^{-\nu_0\gamma t}p^2 + 1/2m \cdot \nu_0\gamma^2(1+\nu_0)e^{\nu_0\gamma t}x^2.$$

$\gamma = 3mc^3/2e^2$ and ν_0 is the absolute value of the negative root of the equation: $\nu^3 - 2\nu^2 + \nu + \omega^2/\gamma^2 = 0$. Again we see that there exists no stationary state at all.

In conclusion, we may say that

(1) the dissipative system has no stationary state,

(2) it has, however, the Hamiltonian function,

(3) the direction of the time axis plays an essential role here, and

(4) the pseudostable orbits in the classical mechanics correspond to the pseudostationary discrete states in the quantum mechanics.

As is well-known, the recent procedures such as the λ-process are the quantization method performed together with the subtraction of the self-field of the electron. However, we may take up the another method of quantization, that is, to quantize the system on the basis of the well-defined classical problem. For example, we may take, as this classical basis, the Maxwell-Dirac theory instead of the Maxwell-Lorentz one, because the latter has no solution even in the classical region. Nevertheless, as was seen above, the results obtained were negative, that is, the stable character of the atomic ground state could not be accounted. Speaking more boldly, it is probably impossible to express adequately the interaction between the electron and its own field by the

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simple dissipative forces alone.

* The full contents of this paper were read at the symposium about the physics held at the Nagoya University, on March 1, 1947.

† P. A. M. Dirac, Proc. Roy. Soc. **167** (1938), 148.

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